GROUP TWO
Data Analysis
2026

Data Analysis

For Better Business Decisions

Superstore Sales

Hello, and Welcome

About The Initiative

Achieving leadership in ICT by building a skilled, innovative workforce that drives comprehensive digital transformation locally and regionally. Implemented in collaboration with leading global, regional, and local tech companies, delivering hands-on workshops, enriching education, and providing training programs to enhance students’ skills.

Team Members

Mohammed Afify

Accountant (10+ years) & Data Analyst (5 years).
Strong belief in data-driven decision-making.
Passionate about dashboards & actionable insights.

Niera sarhan

A fresh statistics graduate, with a solid foundation in data analysis.
Intrested in finding the real why behind the numbers.
Driven to transform raw data into meaningful insights.

Asmaa Ali

Recent Mathematics & Statistics graduate with a strong analytical mindset.
Skilled in data analysis using Python, SQL, Tableau and Excel.
Passionate about turning data into meaningful insights.

Hamada Hassan

Recent Mathematics & Statistics graduate with a strong analytical mindset.
Skilled in data analysis using Python, SQL, and Excel.
Passionate about turning data into meaningful insights.

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Data Analysis
2026

GitHub link

https://github.com/MohamedAfify2025/Data-analysis-project

GROUB TWO

Data Analysis

2026

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Programs Used

- Excel
- S Q L
- python
- Tableau
- Power Bi

Project Documentation Guidelines

01 Data Cleaning and Preprocessing

02 Analysis Questions Phase

03 Dashboard Phase

04 Recommendations

05 Insights and Final Presentation

Data Cleaning and Preprocessing

At this stage, the work was done as follows:

Project Plan

Includes the Time distribution for each step

Task Assignment & Roles

Defines responsibilities for each team member.

Project Proposal

Overview of the project, objectives, and scope.

Cleaning data

to prepare it for analysis

Project Plan – Includes the Time distribution for each step

Task Assignment & Roles – Defines responsibilities for each team member.

1. Import data into SQL &Tableau from the appropriate Excel source.

2. Examine the data to detect missing values, errors, and duplicates.

3. Address missing values through elimination or imputation.

4. Correct data errors (such as spelling errors or illogical values).

5. Transform the data into a format suitable for analysis (standardize formats, extract new variables).

6.Document all processes performed during the cleaning process to ensure transparency and reproducibility.

7. Hand over the cleaned data to the team to move to the analysis phase.

3 Project Proposal

This project aims to analyze Superstore Sales data to identify key factors affecting business performance. It will examine sales by category, product, and region while assessing the impact of shipping methods on customer satisfaction. The analysis will leverage tools like Excel and Power BI to extract insights and recommendations. The expected outcome includes reports and visualizations highlighting trends and strategies to increase profitability and improve operations.

4 Cleaning data -to prepare it for analysis

1. Data Inspection

- Review columns and data types (text, numbers, dates)
- Ensure column names are clear and consistent

2. Handle Missing Values

- Remove unnecessary incomplete rows
- Or replace missing values (0 for numbers / "Unknown" for text)

3. Remove Duplicates

- Use “Remove Duplicates”
- Focus on key fields like Order ID

4 Cleaning data -to prepare it for analysis

4. Standardize Data

- Unify text format (upper/lower case)
- Clean text using functions like TRIM and PROPER

5. Fix Dates and Numbers

- Ensure dates are in correct format
- Verify numbers are clean and free of errors

6. Create Calculated Columns

Example:

- Profit Margin
- Delivery Time

Analysis Questions Phase

General Analysis

1. What are the total sales in each state?

2.Which categories have the highest sales?

3.Which sub-categories achieve the highest sales?

4.What is the average delivery time between the order date and the ship date?

Analysis Questions Phase

Geographical Analysis

1.What are the total sales by region?

2.Which city generate the highest sales?

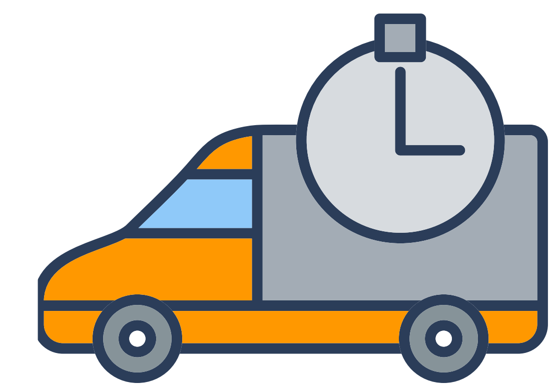
Customer Analysis

1. Who are the top customers based on total sales?

2.Who are the top customers based on order?

Analysis Questions Phase

Time-Based Analysis



1. How are sales distributed monthly or annually?
2. Is there a seasonal sales pattern?
3. Which time period achieves the highest sales?

Analysis Questions Phase

Product Analysis

1.Which products have the highest sales?

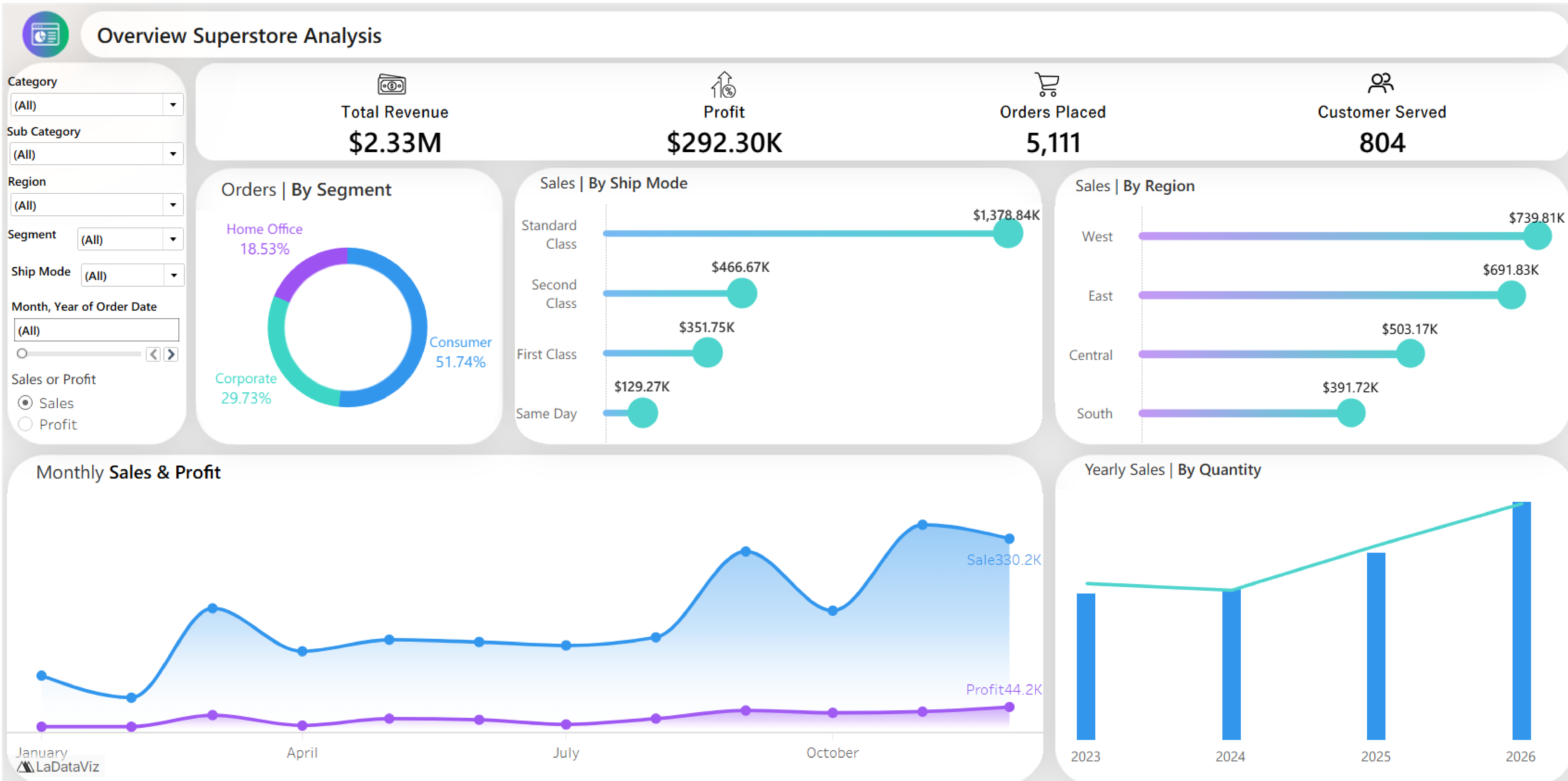
2. Which products have the lowest sales?

Shipping Method Analysis

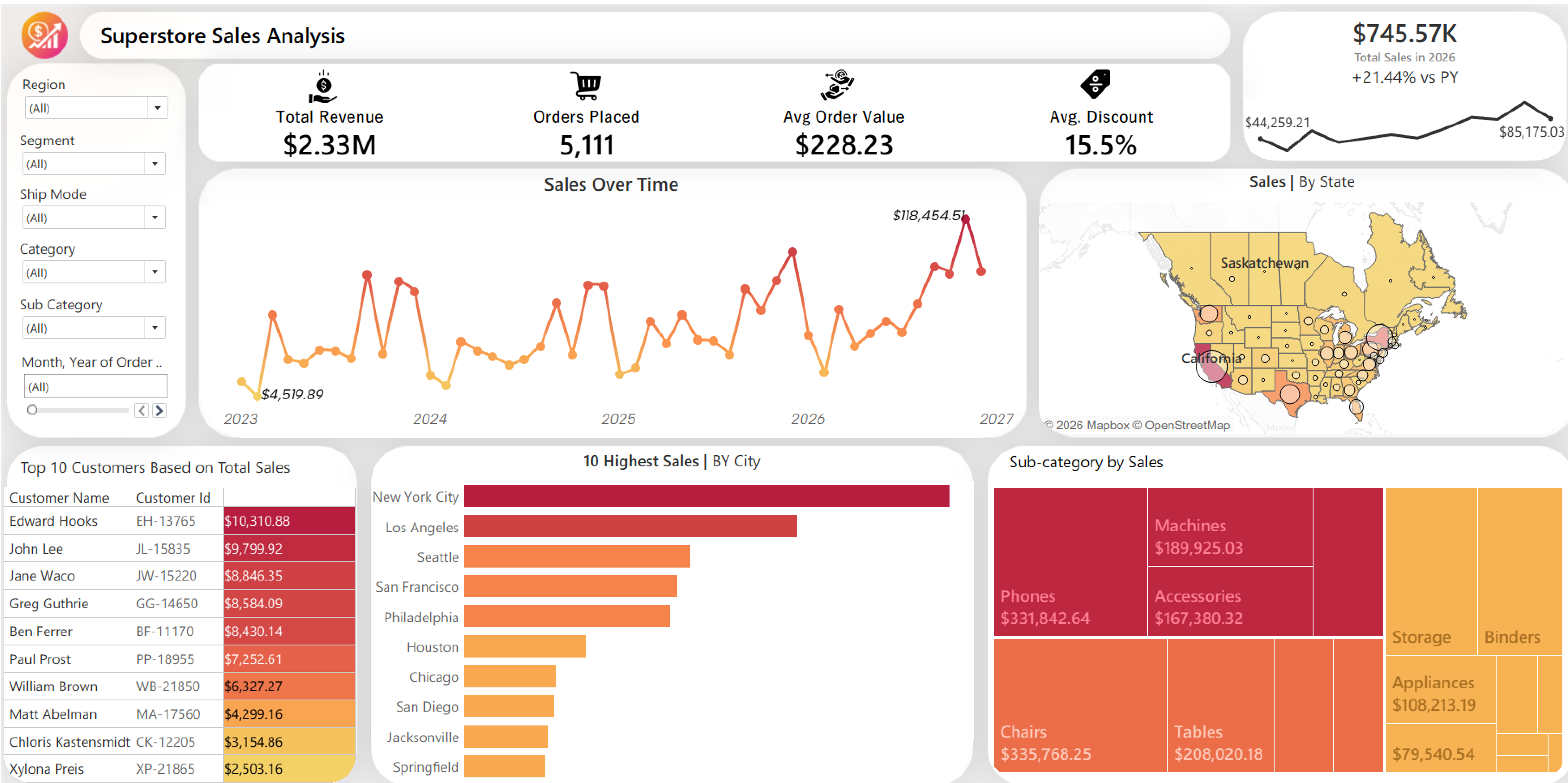
1. Which shipping methods are used the most, and what is their impact on sales?

2. What is the average delivery time for each shipping method?

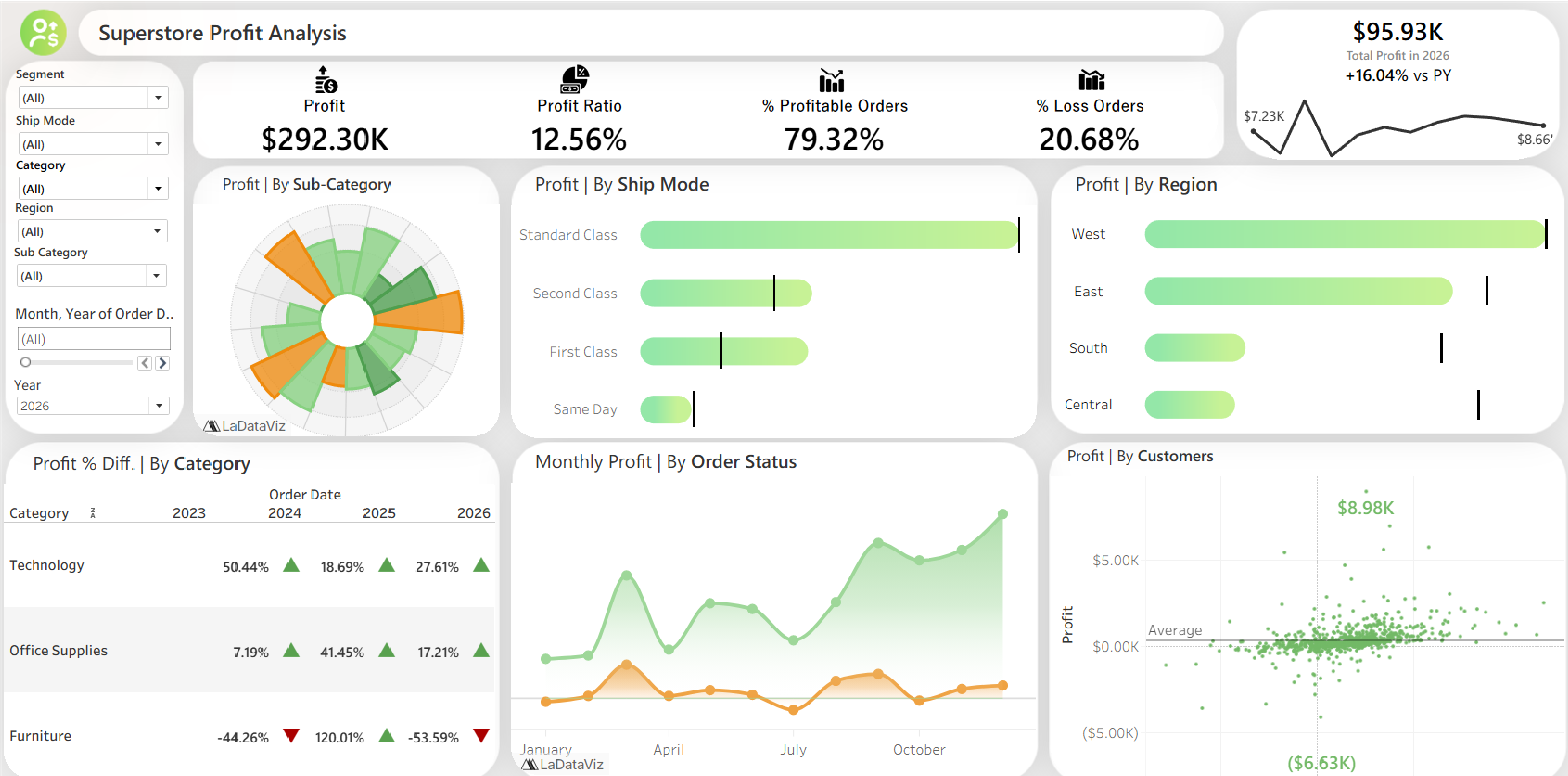
Dashboard Phase



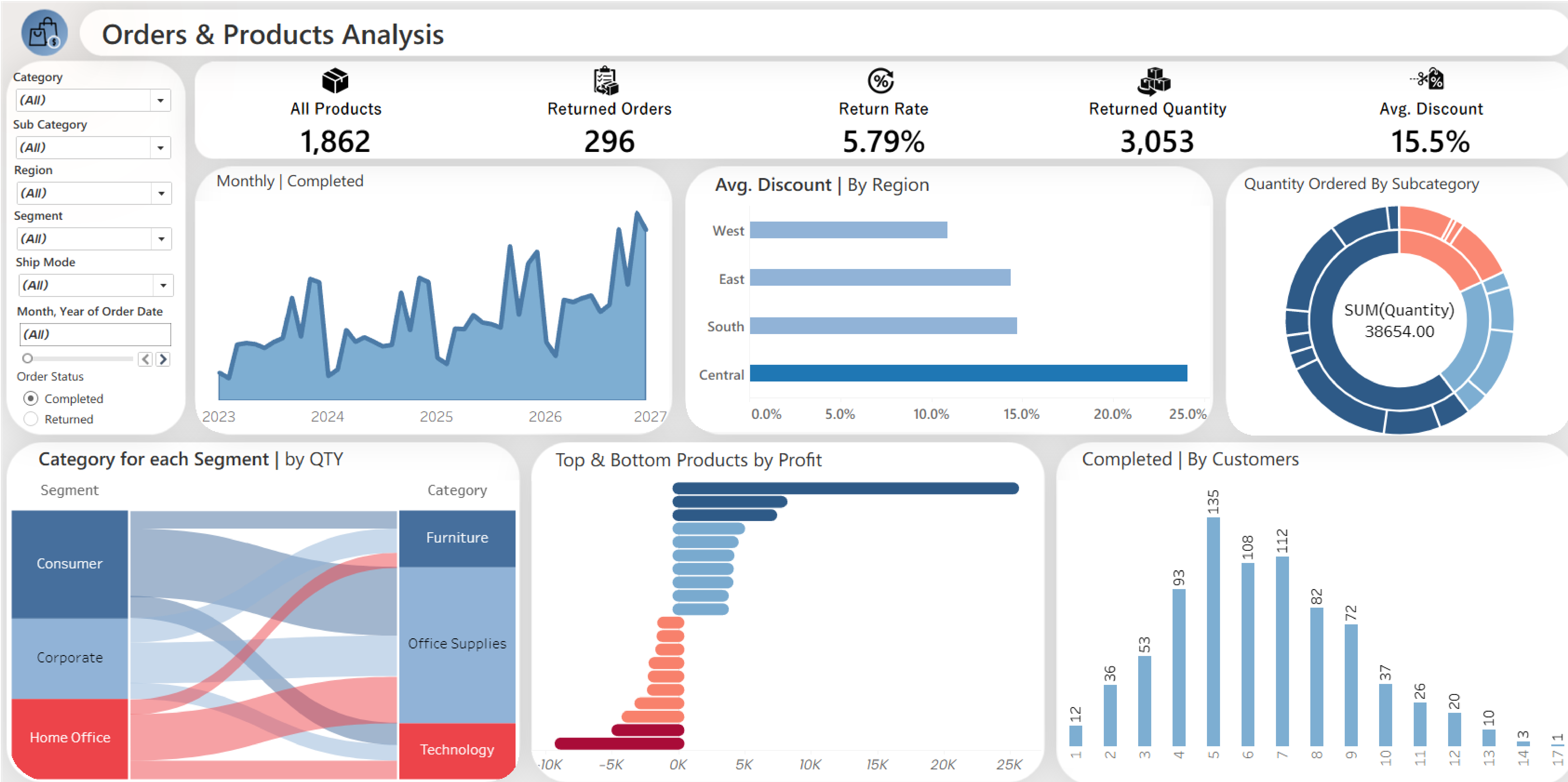
Dashboard Phase



Dashboard Phase



Dashboard Phase

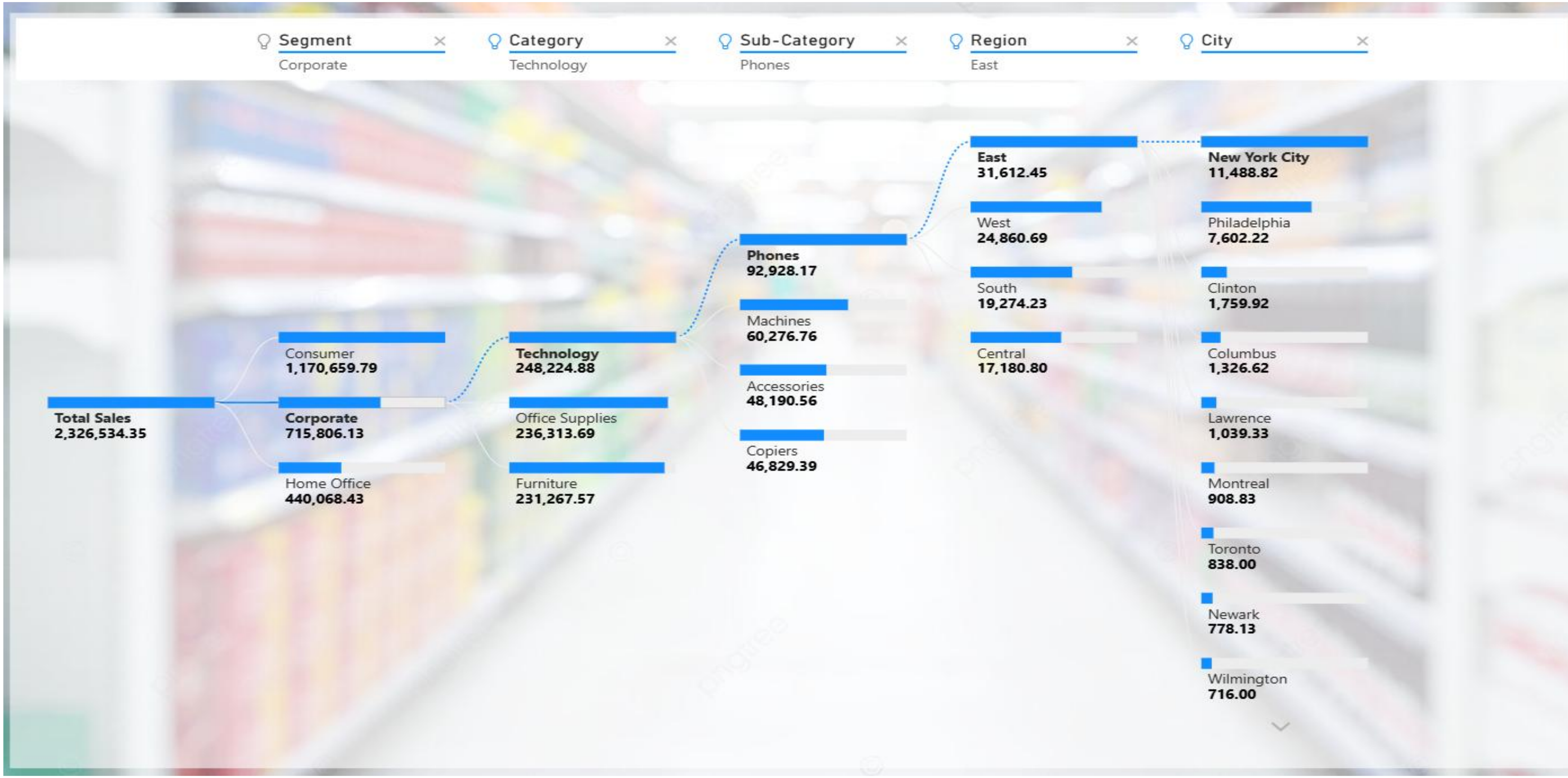


Dashboard Phase

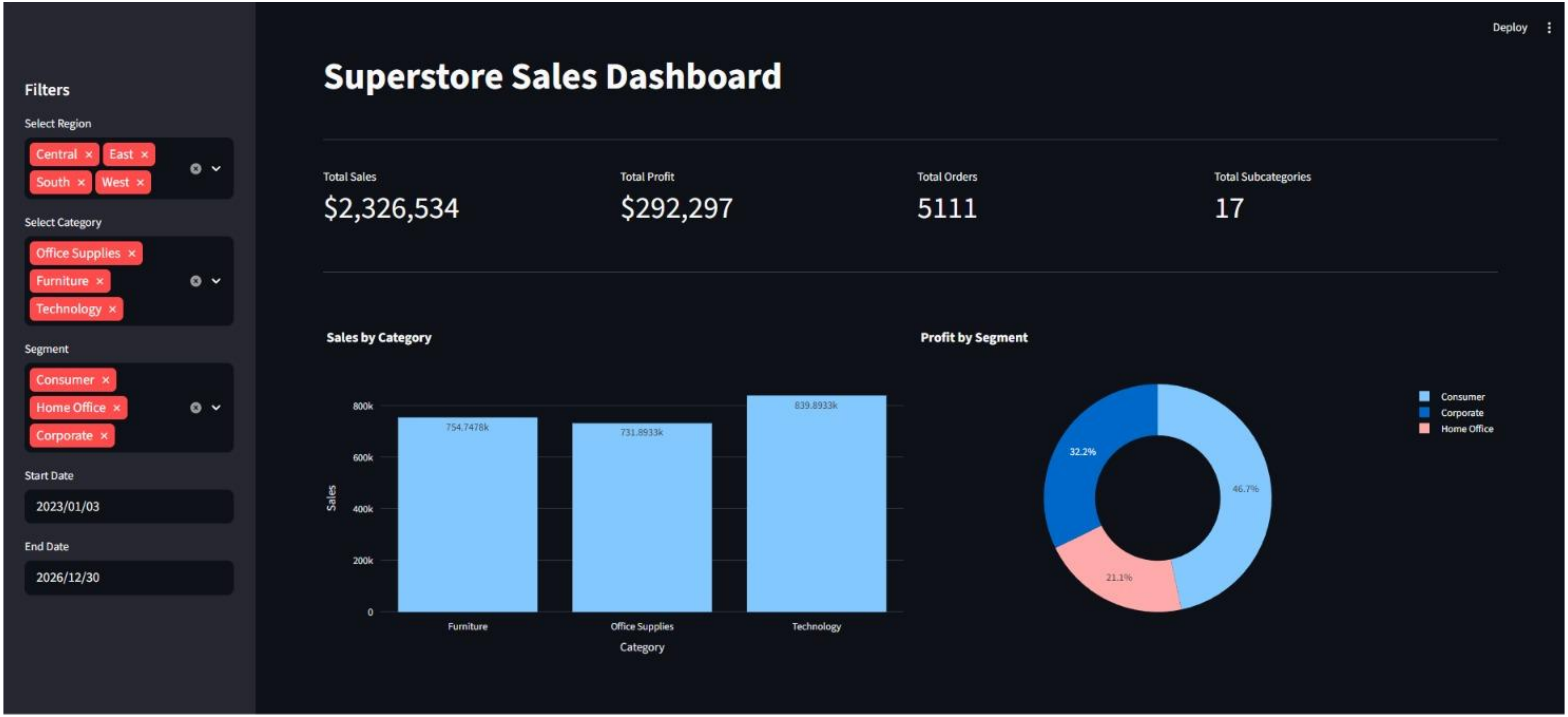
Power BI

Dashboard Phase

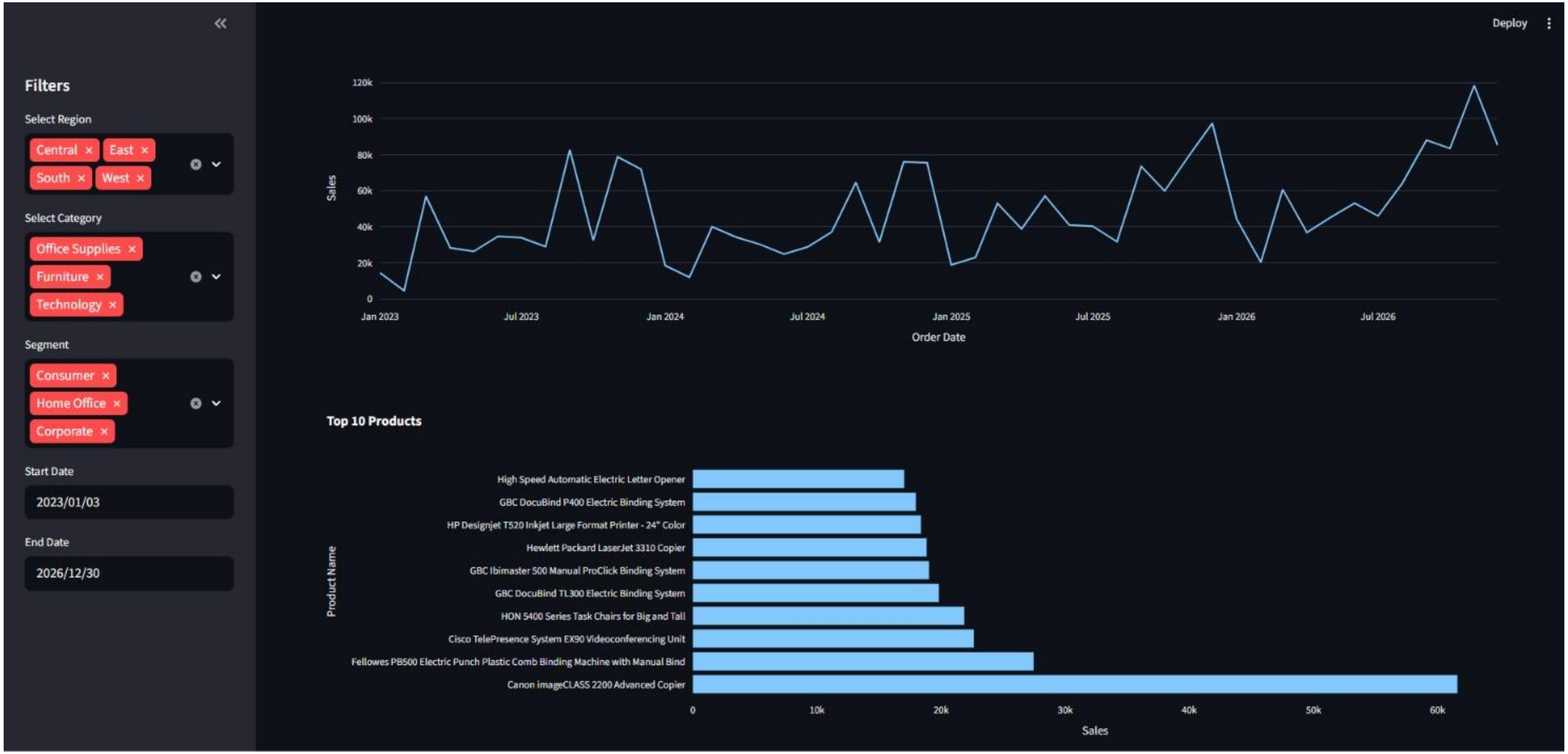
Dashboard Phase



Dashboard Phase



Dashboard Phase



Dashboard Phase

```sql
--1) Overview : How we are going in general ? -----------------------------------
SELECT
Sum(sales) as Total_Sales ,
Sum(profit) as Total_Profit ,
ROUND(Sum(profit) / Sum(sales) * 100 ,2) as Profit_Margin ,
SUM(Quantity) as Total_Units_Sold ,
Count(Distinct Order_ID) as Total_Orders ,
Count(DISTINCT CUSTOMER_ID) as Total_Customers ,
ROUND(AVG([Sales]), 2)   AS Avg_Order_Value
FROM superstore_final
```

100 %   2   0

Results   Messages

	Total_Sales	Total_Profit	Profit_Margin	Total_Units_Sold	Total_Orders	Total_Customers	Avg_Order_Value
1	2326534.34983236	292716.813708261	12.58	38654	5111	804	228.23

Dashboard Phase

```sql
--2) Subcategories and Products -----

---2.1) Which sub-categories are losing money?
SELECT Category,Sub_Category,
COUNT(DISTINCT Order_ID)  AS Total_Orders,
ROUND(SUM(Sales), 2)  AS Total_Sales,
ROUND(SUM(Profit), 2)   AS Total_Profit,
ROUND(SUM(Profit) / (SUM(Sales)) * 100, 2)  AS Profit_Margin_Pct,
COUNT(DISTINCT CASE WHEN Profit < 0 THEN [Order_ID] END)  AS Loss_Orders,
ROUND(AVG(Discount) * 100, 2)  AS Avg_Discount_Pct
FROM superstore_final
GROUP BY Category, Sub_Category
ORDER BY Total_Profit ASC
```

100 % | 2 | 0

Results | Messages

	Category	Sub_Category	Total_Orders	Total_Sales	Total_Profit	Profit_Margin_Pct	Loss_Orders	Avg_Discount_Pct
1	Furniture	Tables	314	208020.18	-17333.21	-8.33	196	25.81
2	Furniture	Bookcases	228	115361.2	-3632.07	-3.15	109	21.53
3	Office Supplies	Supplies	189	46725.5	-1171.39	-2.51	33	7.6
4	Office Supplies	Fasteners	226	8532.24	2428.64	28.46	12	7.86
5	Technology	Machines	114	189925.03	3461.98	1.82	41	30.43
6	Office Supplies	Labels	348	12695.04	5572.78	43.9	1	7.09
7	Office Supplies	Art	756	27659.01	6653.2	24.05	1	7.5
8	Office Supplies	Envelopes	251	16528.36	6988.02	42.28	0	7.97
9	Furniture	Furnishings	919	95598.13	13891.74	14.53	167	13.81
10	Office Supplies	Appliances	459	108213.19	18329.48	16.94	65	16.46
11	Office Supplies	Storage	787	224644.55	21285.11	9.48	154	7.51
12	Furniture	Chairs	591	335768.25	27223.53	8.11	229	16.92
13	Office Supplies	Binders	1339	207354.88	31426.1	15.16	531	36.91
14	Office Supplies	Paper	1205	79540.54	34511.51	43.39	1	7.5
15	Technology	Accessories	718	167380.32	41936.64	25.05	86	7.85
16	Technology	Phones	826	331842.64	45050.83	13.58	126	15.26
17	Technology	Copiers	70	150745.29	56093.94	37.21	0	15.71

Dashboard Phase

```sql
--2.2) Top 10 products losing the most money
SELECT TOP 10 Product_Name,Category,Sub_Category,
COUNT(DISTINCT Order_ID) AS Total_Orders,
ROUND(SUM(Sales), 2) AS Total_Sales,
ROUND(SUM(Profit), 2) AS Total_Profit,
ROUND(AVG(Discount) * 100, 2) AS Avg_Discount_Pct
FROM superstore_final
GROUP BY Product_Name, Category, Sub_Category
ORDER BY Total_Profit ASC
```

Results | Messages

	Product_Name	Category	Sub_Category	Total_Orders	Total_Sales	Total_Profit	Avg_Discount_Pct
1	Cubify CubeX 3D Printer Double Head Print	Technology	Machines	3	11099.96	-8879.97	53.33
2	Lexmark MX611dhe Monochrome Laser Printer	Technology	Machines	4	16829.9	-4589.97	40
3	Cubify CubeX 3D Printer Triple Head Print	Technology	Machines	1	7999.98	-3839.99	50
4	Chromcraft Bull-Nose Wood Oval Conference Tables ...	Furniture	Tables	5	9917.64	-2876.12	28
5	Bush Advantage Collection Racetrack Conference Table	Furniture	Tables	7	9544.72	-1934.4	35
6	GBC DocuBind P400 Electric Binding System	Office Supplies	Binders	6	17965.07	-1878.17	45
7	Cisco TelePresence System EX90 Videoconferencing ...	Technology	Machines	1	22638.48	-1811.08	50
8	Martin Yale Chadless Opener Electric Letter Opener	Office Supplies	Supplies	6	16656.2	-1299.18	10
9	Balt Solid Wood Round Tables	Furniture	Tables	4	6518.75	-1201.06	20
10	Riverside Furniture Oval Coffee Table, Oval End Table,...	Furniture	Tables	5	4446.18	-1147.4	30

Dashboard Phase

```sql
--2.3) Top 10 products by profit--

SELECT TOP 10 product_name, category,sub_category,
COUNT(DISTINCT Order_ID) AS Total_Orders,
ROUND(SUM(profit), 2) AS Total_profit,
ROUND(SUM(Sales), 2) AS Total_Sales,
ROUND(AVG(Discount) * 100, 2) AS Avg_Discount_Pct
FROM superstore_final
GROUP BY product_name,Category,Sub_Category
ORDER BY Total_profit DESC
```

100 % 2 0

Results Messages

	product_name	category	sub_category	Total_Orders	Total_profit	Total_Sales	Avg_Discount_Pct
1	Canon imageCLASS 2200 Advanced Copier	Technology	Copiers	5	25199.93	61599.82	12
2	Fellowes PB500 Electric Punch Plastic Comb Bindin...	Office Supplies	Binders	10	7753.04	27453.38	24
3	Hewlett Packard LaserJet 3310 Copier	Technology	Copiers	8	6983.88	18839.69	20
4	Canon PC1060 Personal Laser Copier	Technology	Copiers	4	4570.93	11619.83	15
5	HP Designjet T520 Inkjet Large Format Printer - 24" ...	Technology	Machines	3	4094.98	18374.9	16.67
6	Ativa V4110MDD Micro-Cut Shredder	Technology	Machines	2	3772.95	7699.89	0
7	3D Systems Cube Printer, 2nd Generation, Magenta	Technology	Machines	2	3717.97	14299.89	0
8	Plantronics Savi W720 Multi-Device Wireless Heads...	Technology	Accessories	7	3696.28	9367.29	5.71
9	Ibico EPK-21 Electric Binding System	Office Supplies	Binders	3	3345.28	15875.92	33.33
10	Zebra ZM400 Thermal Label Printer	Technology	Machines	2	3343.54	6965.7	0

Dashboard Phase

```sql
--3 D I S C O U N T ------------------------

SELECT ROUND(Discount * 100, 0)AS Discount_Pct,
COUNT(DISTINCT Order_ID)AS Orders,
ROUND(SUM([Profit]), 2)AS Total_Profit,
ROUND(SUM([Sales]), 2) AS Total_Sales,
ROUND(AVG(Profit), 2)AS Avg_Profit,
ROUND(AVG(Sales), 2)AS Avg_Sales
FROM superstore_final
GROUP BY ROUND([Discount] * 100, 0)
ORDER BY Discount_Pct
```

	Discount_Pct	Orders	Total_Profit	Total_Sales	Avg_Profit	Avg_Sales
1	0	2714	326718.59	1105323.79	66.34	224.43
2	10	91	9099.97	54952.5	94.79	572.42
3	15	51	1418.99	27558.52	27.29	529.97
4	20	2440	91079.95	773939.39	24.58	208.83
5	30	214	-10513.45	104474.07	-45.71	454.24
6	32	27	-2391.14	14493.46	-88.56	536.79
7	40	186	-23086.37	116497.76	-111.53	562.79
8	45	10	-2073.11	5484.97	-207.31	498.63
9	50	64	-20506.43	58918.54	-310.7	892.71
10	60	138	-6164.35	7105.94	-41.37	47.69
11	70	349	-40300.57	40805.27	-95.05	96.24
12	80	251	-30565.27	16980.15	-101.55	56.41

Dashboard Phase

```sql
---4 R E T U R N S ----

--4.1) Returns by category and subcategory--------------------
SELECT Category,Sub_Category,
COUNT(DISTINCT [Order_ID]) AS Total_Orders,
COUNT(DISTINCT CASE WHEN Returned = 'Yes' THEN Order_ID END) AS Returned_Orders,
ROUND(100.0 * COUNT(DISTINCT CASE WHEN Returned = 'Yes' THEN Order_ID END) / (COUNT(DISTINCT Order_ID) ), 2) AS Return_Rate_Pct,
SUM(CASE WHEN Returned = 'Yes' THEN Quantity ELSE 0 END) AS Returned_Quantity,
ROUND(SUM(CASE WHEN Returned = 'Yes' THEN Sales ELSE 0 END), 2) AS Returned_Sales_Value,
ROUND(SUM(CASE WHEN Returned = 'Yes' THEN Profit ELSE 0 END), 2) AS Profit_Lost_From_Returns
FROM superstore_final
GROUP BY Category, Sub_Category
ORDER BY Return_Rate_Pct DESC

--4.2) Returns by customer segment---------------------------
```

100 % 2 0 Ln: 10

Results Messages

	Category	Sub_Category	Total_Orders	Returned_Orders	Return_Rate_Pct	Returned_Quantity	Returned_Sales_Value	Profit_Lost_From_Returns
1	Technology	Machines	114	13	11.400000000000	61	13158.39	-262.73
2	Furniture	Tables	314	30	9.550000000000	134	17042.28	-1058.08
3	Office Supplies	Appliances	459	40	8.710000000000	180	10129.78	2571.67
4	Technology	Phones	826	71	8.600000000000	263	27633.61	2780.91
5	Office Supplies	Supplies	189	16	8.470000000000	62	3195.96	120.66
6	Office Supplies	Fasteners	226	19	8.410000000000	68	213.09	91.74
7	Office Supplies	Paper	1205	99	8.220000000000	453	7140.28	3237.1
8	Office Supplies	Binders	1339	108	8.070000000000	558	9983.15	-1802.28
9	Furniture	Chairs	591	47	7.950000000000	177	25154.7	1207.02
10	Furniture	Furnishings	919	67	7.290000000000	277	9577.55	2077.88
11	Technology	Accessories	718	52	7.240000000000	221	12716.33	2868.23
12	Technology	Copiers	70	5	7.140000000000	19	19199.84	8610.94
13	Office Supplies	Storage	787	53	6.730000000000	211	14679.93	1504.4
14	Furniture	Bookcases	228	15	6.580000000000	66	7444.64	114.3
15	Office Supplies	Art	756	46	6.080000000000	166	1429.12	349.54
16	Office Supplies	Labels	348	20	5.750000000000	87	885.87	404.95
17	Office Supplies	Envelopes	251	13	5.180000000000	50	919.75	416.13

Dashboard Phase

```sql
--4.2) Returns by customer segment----------------------------
SELECT Segment,
COUNT(DISTINCT Order_ID)  AS Total_Orders,
COUNT(DISTINCT CASE WHEN Returned = 'Yes' THEN Order_ID END) AS Returned_Orders,
ROUND( 100.0 *  COUNT(DISTINCT CASE WHEN Returned = 'Yes'
THEN Order_ID END) / NULLIF(COUNT(DISTINCT Order_ID), 0), 2)  AS Return_Rate_Pct,
ROUND(SUM(CASE WHEN Returned = 'Yes' THEN Profit ELSE 0 END), 2) AS Profit_Lost
FROM superstore_final
GROUP BY Segment
ORDER BY Return_Rate_Pct DESC
```

100 % | 2 | 0

Results | Messages

	Segment	Total_Orders	Returned_Orders	Return_Rate_Pct	Profit_Lost
1	Corporate	1552	93	5.990000000000	5452.42
2	Consumer	2628	154	5.860000000000	17002.19
3	Home Office	931	49	5.260000000000	777.74

Dashboard Phase

```sql
--4.3) Returns by Region and state-----------------
SELECT region,[Regional Manager],
State_Province, COUNT(returned) AS Returned_Orders,
count(distinct Order_ID) as total_orders,
ROUND( 100.0 *  COUNT(DISTINCT CASE WHEN Returned = 'Yes'
THEN Order_ID END) / NULLIF(COUNT(DISTINCT Order_ID), 0), 2)  AS Return_Rate_Pct,
round(sum(sales),2)as total_sales
FROM superstore_final
GROUP BY region,State_Province,[Regional Manager]
ORDER BY Return_Rate_Pct DESC
```

100 %   1   0

Results   Messages

	region	Regional Manager	State_Province	Returned_Orders	total_orders	Return_Rate_Pct	total_sales
1	West	Sadie Pawthorne	Utah	12	26	15.380000000000	11220.06
2	West	Sadie Pawthorne	Montana	2	8	12.500000000000	5589.35
3	West	Sadie Pawthorne	Oregon	20	56	12.500000000000	17431.15
4	West	Sadie Pawthorne	California	338	1021	12.440000000000	457687.63
5	West	Sadie Pawthorne	Colorado	25	79	11.390000000000	32108.12
6	West	Sadie Pawthorne	Washington	66	256	11.330000000000	138641.27
7	West	Sadie Pawthorne	New Mexico	3	22	9.090000000000	4783.52
8	West	Sadie Pawthorne	Idaho	2	11	9.090000000000	4382.49
9	East	Chuck Magee	Delaware	10	44	9.090000000000	27451.07
10	West	Sadie Pawthorne	Arizona	22	108	8.330000000000	35282

Key points of the data

- **The data contains over 10,000 records, making it suitable for analysis and pattern extraction.**
- **There are no missing values or clearly duplicate records within the file.**
- **The Technology department generates the highest sales revenue compared to other departments.**
- **The West region generates the highest profits, while the Central region is the least profitable.**

Key points of the data

A significant number of transactions show negative profitability, indicating losses in some sales.

- **Key examples of negative profit (loss) transactions:**

- **Some orders in the Furniture category, particularly Tables, achieved high sales but negative profits due to large discounts.**
- **Chairs were sold at a cost higher than the expected profit margin, resulting in losses.**
- **Some orders in the Central region recorded recurring losses compared to other regions.**
- **High discounts on some invoices turned profits into losses.**
- **Some low-priced products with high shipping costs negatively impacted net profit.**

Recommendations

- **Analyze the causes of unprofitable operations and reduce high discounts.**
- **Focus on higher-profitability products and categories to increase revenue.**
- **Create a dashboard to track sales and profits by region and category.**
- **Study the performance of low-profitability regions and improve sales strategies there.**
- **Use predictive analytics to forecast sales and improve inventory management.**

Thank You!

FOR YOUR ATTENTION